Fragment-Resolved Nonlocal Correlation: Pair Attribution of the VV10 Term

Angran Xia*
School of Chemistry & Chemical Engineering, Shaanxi Normal University, Xi'an 710119, Shaanxi, China
haangyeon@snnu.edu.cn

Changwei Wang
School of Chemistry & Chemical Engineering, Shaanxi Normal University, Xi'an 710119, Shaanxi, China

Draft — 29 September 2026
ORCIDs PENDING, author-supplied

Authors. Angran Xia (corresponding; haangyeon@snnu.edu.cn) and Changwei Wang — School of Chemistry & Chemical Engineering, Shaanxi Normal University, Chang'an Campus, Xi'an 710119, Shaanxi, China. Both names, the institution and the campus postcode were supplied by the author on 2026-09-30; the school's English wording is taken from the school's own English site rather than translated by us. ORCIDs and the second author's e-mail are PENDING — author-supplied, not guessed. The Chinese forms of both names are recorded in `CITATION.cff` and in `paper/software.md` and are deliberately not typeset here: this file goes through a pdflatex with no CJK font, and `tools/md2tex.js` refuses characters it cannot map rather than dropping them silently.

Number provenance is tagged `[S#]`; the tag-to-artefact table is the appendix at the end of this file, mirrored in `paper/tex_audit.md`. The literature search behind the no-precedent claim is described where its own limits are stated (§1 and §6).

1 Abstract

The VV10 nonlocal-correlation term [1] of range-separated hybrid functionals is a double sum over pairs of grid points, and production implementations fold that sum into per-point values before returning them, which destroys the pair index. We evaluate the double sum explicitly, keep the (i, j) index, and attribute each grid point to atoms and fragments through a real-space partition, so the term resolves into intra-fragment blocks and inter-fragment pair blocks. The split is exact rather than approximate: the term's local part is beta times the electron number and cancels identically in any superposition difference, so decomposing the double sum alone loses nothing. On a fixed density the explicit sum reproduces PySCF's own kernel to 5.7e-06 kcal/mol, and the superposition differences of that term agree with ORCA 6.0.1's independently computed `NL Energy, E(C,NL)` to under 1% of the term for all eight complexes examined.

Applied to eight S22 complexes at official GMTKN55 geometries (omegaB97X-V/6-31G*), the decomposition argues against a common shorthand. The nonlocal and semi-local parts of the exchange-correlation change are not locked in a fixed ratio: the nonlocal term supplies 16% of it

*Corresponding author; e-mail in the byline.

for the water dimer but 97% and 52% for the two structures of the same benzene dimer, and on the methane dimer the two parts carry opposite signs, so no share is quoted there and the nonlocal term alone (-0.517 kcal/mol) exceeds the counterpoise-corrected interaction energy (-0.466). Association-time nonlocal correlation is carried almost entirely by point pairs that straddle the fragments: the inter-fragment block accounts for 0.92-1.08 of the frozen-density superposition difference in the six systems checked against a counterpoise reference. What does not collapse is the choice of attribution scheme - 0.004-0.247 kcal/mol absolute and 0.8-18.2% of the Becke-scheme value, stable across bases and grids but with no pattern by interaction type - so a fragment-resolved nonlocal energy needs a per-system error bar, not a global one. For absolute rather than differential fragment energies that scheme dependence is exactly beta times the dependence of the Becke population, 3.04 kcal/mol per electron, and we report it as a limitation rather than a result.

2 1. Introduction

Range-separated hybrid functionals with the VV10 nonlocal-correlation functional — ωB97X-V [2] and ωB97M-V [3] in particular [4] — have become a default choice for non-covalent interactions. The nonlocal term is a two-electron-density double integral over all pairs of grid points,

E_NL = $\frac{1}{2}$ $\int\int$ ρ(r) ρ(r′) Φ(r, r′) d^3r d^3r',

and every production quantum-chemistry package we have examined evaluates it in a way that destroys the pair index: PySCF's `nr_nlc_vxc` calls its own C kernel `libdft.VXC_vv10nlc`, which bins the kernel and returns only point-folded values. We have not audited the equivalent paths inside libxc [5], ORCA, Psi4 or Gaussian - libxc is not one of them and cannot be made into a numerical referee: the two libxc shared objects on this machine (conda-forge 5.2.3 and the 7.0.0 bundled with PySCF) export `xc_nlc_coef` and nothing else among thirteen probed nonlocal entry points - no `xc_nlc_vxc`, no `xc_run` - which matches libxc's own statement that this class of van der Waals term "cannot be evaluated with the local density information provided to Libxc, and must be handled by the calling program". libxc therefore supplies the parameters (b, C) keyed by functional name, not an independent value of the integral, so an agreement claim against it would be a category error (see §4, `RESEARCH-20260929-lit.md` §8.3, and `evidence/recompute/20260929T110830Z_libxc_referee.out` for the symbol audit). PySCF's nonlocal term comes from its own `nr_nlc_vxc` implementation, not from libxc. The consequence we do demonstrate is structural rather than numerical: once the double sum has been folded into a per-point energy density, as ours does not, the question "how much of this nonlocal correlation belongs to the interaction between fragment A and fragment B?" has no answer inside that evaluation.

We provide that answer. `nlcsplit` evaluates the same double sum explicitly, keeping the (i, j) pair index, and attributes each grid point to a fragment through a real-space partition, so that E_NL decomposes into intra-fragment blocks E_AA and inter-fragment pair blocks E_AB. Evaluating the sum explicitly is not itself new: JuNoLo post-processed a converged density from another code with the nonlocal term, at the same N^2 cost, in order to obtain an improved *total* energy on massively parallel hardware [S14]. What is absent from that work - and from the production kernels - is the attribution. JuNoLo in fact recommends against fragment-style treatments for the weakly-bound cases where fragments are distinguishable, reserving itself for chemisorption "where one cannot tell apart fragments of the bonded system" [S14]; the present paper works precisely in the regime where they are. The decomposition is exact by construction — the sum of the blocks reproduces the total to machine precision — but exactness is not the point and we do not sell it

as such: grouping the summation indices is trivially additive. Two things are not trivial, and they are the content of this paper.

The first is **attribution**. The non-uniqueness lives in which fragment owns each grid point (and, in the same spirit, how the local term $\beta\int\rho$ is assigned), not in how the sum is grouped. We therefore quantify the spread over four published real-space partitions — Becke, Becke with the radius adjustment, Becke–Treutler, and Stratmann's — on eight S22 complexes, and report it as a per-system uncertainty rather than asserting one scheme as correct. The spread is 0.004–0.247 kcal/mol in absolute terms and 0.8–18.2% relative to |E_AB|, and it shows no systematic ordering by interaction type: the benzene sandwich sits at 0.8% while benzene–water sits at 18.2% [S2]. Nor does a better one-electron basis remove it: on the official water-dimer geometry the spread stays 0.041-0.044 kcal/mol across the four basis sets beyond STO-3G (0.030 in STO-3G itself, where the quantity is smaller too) while the quantity itself drifts by 0.18 kcal/mol along the same ladder (section 3.4, [S12]). A single blanket uncertainty figure would misrepresent the data; we report it system by system.

The second, and chemically the more consequential, is **what the term actually is**. Within a hybrid functional the exchange-correlation contribution to an interaction energy splits as ΔE_xc = ΔE_xc$\hat{}$rest + ΔE_NLC, where "rest" collects the semi-local exchange-correlation and the fraction of exact exchange, and the local piece $\beta\int\rho$ drops out of any superposition difference identically — β is a constant and $\int\rho$ equals the electron count, so $\Delta\beta\int\rho$ vanishes to 3.7×10^{-5} kcal/mol in our level-1 test [S3]. That leaves the double sum as the only part of the nonlocal term that can carry inter-fragment structure, and measuring it shows that treating it as "the dispersion" is unsafe. For the water dimer it supplies 16% of ΔE_xc. For the two benzene dimer configurations of S22 the same quantity ranges from 52% to 97% [S4] — and those two structures differ only in the relative orientation of the same pair of molecules, so the failure of the identification is visible within one chemical system, without any cross-system comparison. In the methane dimer the nonlocal term exceeds the whole of ΔE_xc (ΔE_NLC $= -0.517$ against ΔE_xc $= -0.232$), i.e. the rest of the exchange-correlation is net repulsive there.

Our positioning relative to the closest prior work is deliberate, and it is narrower than the title suggests. Dasgupta and Paesani [6] decompose interaction energies of dispersion-corrected functionals with second-generation ALMO-EDA (E_Int = E_Pol + E_Frz + E_CT + E_Disp) and obtain E_Disp as a difference against a dispersion-free reference: they treat the VV10 correction as one undivided block and ask whether that block is accurate. We work inside that block. Heßelmann [7] partitions his own nonlocal scheme's correlation energy into atom-atom contributions - the abstract of that paper says so in as many words - so attributing a nonlocal correlation term to atoms is not ours and is not claimed here. JuNoLo [8] [S14] evaluates the nonlocal double sum explicitly, at the same N^2 cost, but post-processes a converged density into an improved *total* energy, and recommends itself for bonded systems "where one cannot tell apart fragments", directing separable cases elsewhere. Hyldgaard, Berland and Schröder [9] read the same quantity as a coupling of zero-point energies of semilocal exchange-correlation holes: a real-space *interpretation*, and the strongest available objection to calling the term unanalyzable, so we keep the distinction explicit - they interpret the kernel and the holes, we take an already-evaluated E_NLC and *bookkeep* it against named fragments and atom pairs, then report how much the bookkeeping itself moves the answer.

What survives that comparison is specific: taking an existing, already-parameterised VV10 double sum on a numerical integration grid, resolving it into named fragment and atom pairs without changing the total, and measuring how much the attribution itself - not the energy - moves when the partition scheme is changed. A formal Web of Science search over the Core Collection (58,664,816 records, 2026-09-29) found no precedent for that [S16]: the two strings written closest to

our own claim both return zero records, and the non-zero strings were enumerated in full through the database's own record endpoint, so every returned record was read rather than the top one. The nearest neighbours are each attributable in one clause. Kruse, Banáš and Sponer [10] benchmark VV10 among 29 functionals and decompose the *total* interaction energy by SAPT; Hujo and Grimme [11] benchmark VV10 for hydrogen bonds; Sinha and Pavanello [12] construct a *new* subsystem-based nonlocal functional rather than resolving an existing one; Hamada and Tsukada [13] attribute an *adsorption energy* to the term, the opposite direction; Vydrov and Van Voorhis [14] re-derive the nonlocal construction itself; and two further result lists of 33 records each - one spanning the whole "van der Waals density functional" [15] x "nonlocal correlation" topic, one crossing "nonlocal correlation" with the partition schemes themselves (Becke, Hirshfeld, Mulliken, atoms-in-molecules) - contain, on reading every title, no application of an atomic partition to the VV10 double sum.

Two cautions about that instrument belong with the claim, because a reader who does not know them will over-read the zeros. A keyword-shaped novelty query is narrower than the claim it is meant to test: the first round of the search [S1] covered the decomposition vocabulary and missed Hyldgaard et al. entirely, a paper whose abstract shares no term with any of our keywords, and the broadest topical query available is what surfaced it. And the open indexes tried first - OpenAlex and Crossref - retrieve a known title reliably when the query resembles its title but miss that same exactly-on-topic paper for legitimate topical phrasings, at 0/4 each [S15]; they are known-item lookup tools here, not topical search, and their zeros are not read as absence. The calibration behind both statements, and the single-index coverage limit of the search, are in section 4.4.

Two limits are stated here rather than in a closing section, because both bound what the numbers may be used for. First, this is not an energy decomposition analysis: we do not separate electrostatics, exchange-repulsion, polarization or charge transfer. Counterpoise-corrected interaction energies are reported for six of the eight systems (§3.2), so basis-set superposition error is measured rather than assumed away, but the correction is not carried through the remaining two benzene-dimer geometries. Second, the explicit sum is not more accurate than the production kernels it is compared against — the interpolation residual of PySCF's native kernel is of order 4×10^{-6} kcal/mol, far below any chemical interest, and the ≈0.1–0.9 kcal/mol offsets that an earlier version of this work attributed to interpolation were in fact a pairing error in our own kernel. The value of the explicit implementation is that it retains the pair index, nothing more.

3 2. Method

3.1 2.1 The VV10 kernel and the pairing convention

With ρ the electron density and $\sigma = |\nabla\rho|^2$,

w0(r) = $\sqrt{}$(C σ^2/ρ^4 + $(4\pi/3)$ ρ), κ(r) = b $(3\pi/2)$ $(\rho/9\pi)$^1/6, g = w0(r_i) R^2 + κ(r_i), g′ = w0(r_j) R^2 + κ(r_j), $\Phi = -3$ / (2 g g′ (g+g′)),

with (b, C) = (6.0, 0.01) for ωB97X-V, and E_NLC = E_NL + $\beta\int\rho$, $\beta = (1/32)(3/b^2)$^3/4. The pairing of ω and κ in g and g′ is a genuine choice, and the cross pairing g = w0(r_j) R^2 + κ(r_i) — which an earlier implementation of ours used — is wrong: at grid level 0 it raises monomer energies by +0.138 kcal/mol for water, +0.418 for argon and +0.906 for benzene, and the offset is itself grid-dependent, so it cannot be subtracted as a constant [S1, S5]. The same-centre pairing above reproduces PySCF's native kernel to 5.7×10^{-6} kcal/mol on the water monomer, and agrees with an independently written script that evaluates the same quantity by a different code path to 1.24×10^{-4} kcal/mol worst case: against the script's self-consistent column the difference runs from -1.058×10^{-4} (level 0) to -1.242×10^{-4} (level 8) and back to -1.230×10^{-4} (level 9), same sign at all ten levels and no drift with grid density (level 9 carries 430,768 integration points, of which 428,920

survive the density floor); its fixed-density column differs only at level 0 (-1.056×10^{-3}), where that column is itself the outlier [S21]. A second program has since been brought to bear on the same quantity - ORCA 6.0.1, on all eight systems, with every residual below 1% of the term - but its grids are not matched to ours, so that anchor is quoted as a joint grid-plus-kernel difference in §4 rather than as a kernel-only check here.

3.2 2.2 Blocks, half-blocks and additivity

`pair_energy(A, B)` returns the half block $\frac{1}{2}$ Σ_i∈A Σ_j∈B w_i w_j Φ_ij, so that the inter-fragment term reported by `fragment_decomposition` is E_AB = 2·pair_energy(A,B) and

E_NL = Σ_A E_AA + Σ_A<B E_AB

holds by construction. We record the additivity residual rather than trusting it: across all systems and partitions reported here it is 0.00–6.94×10^{-18} Ha [S2, S6]. R^2 is evaluated through the polarisation identity $|a|^2+|b|^2-2a\cdot b$ in blocked gemm operations rather than through an explicit displacement tensor, which reduces peak memory by roughly a factor of five at this system size.

3.3 2.3 Attribution schemes and the far-field cut

Grid points are assigned to atoms by PySCF's own partition machinery (`gen_grid.get_partition` with `radi.BRAGG_RADII`), in four variants — Becke [16], Becke with the radius adjustment, Becke–Treutler, Stratmann [17] — plus a nearest- atom hard partition on the common grid as a control. Points with ρ below `rho_floor` (default 10^{-12}) are excluded from the double sum; across the sweep this drops 10.7–22.6% of the points, and the points it drops carry at most 2.3×10^{-6} electrons. Over the floor interval the sweep accepts (10^{-10} to 10^{-14}) E_AB moves by 2.1×10^{-11} Ha at grid level 1 and 4.2×10^{-11} Ha at level 2; adding the two outer rungs at 10^{-8} and 10^{-16} still leaves the reported value unchanged at six decimals in kcal/mol [S7]. The local term $\beta\int\rho$ is a one-body functional of the density and is not pair-resolved; as noted above it cancels identically from superposition differences, which is what makes it legitimate to decompose only the double sum.

3.4 2.4 Geometries and fragment boundaries

All dimer geometries are taken verbatim from the official GMTKN55 repository (github.com/grimme-lab/GMTKN55, `S22/<NN>/struc.xyz`), fetched by `nlcsplit/tools/get_s22.sh`; the repository's own citation instruction is Goerigk et al. [18], PCCP 19, 32184-32215 (2017), DOI 10.1039/C7CP04913G, which we verified by reverse DOI lookup. The S22 database itself is Jurečka et al., PCCP 8, 1985-1993 (2006) [19], also confirmed by reverse DOI lookup. We do not trace these twenty-two complexes back beyond that: an attribution to a still earlier hydrogen-bonded benchmark (PCCP 5, 1811 (2003)) resolves to no Crossref record on two separate attempts (2026-09-29 and 2026-09-30, queries archived in `evidence/bib_provenance/G_s22_provenance_20260930.md`), so we cite only the chain that could be pinned. The geometry files we used come from the GMTKN55 distribution, so our numbers depend on that, not on any earlier compilation. Fragment boundaries are derived from covalent connectivity with a 1.65 Å cutoff — longer than the longest covalent bond appearing in these systems (aromatic C–C, 1.397 Å; a general C–C single bond is 1.54 Å) and shorter than the shortest hydrogen bond (O···H 1.95 Å) — and the code refuses to proceed unless exactly two fragments result. This is not a formality: in the official S22 water-dimer file the atoms are ordered so that the fragments interleave, and a hard-coded index split would mix the two molecules silently. For the water dimer we additionally report ωB97X-V/6-31G* optimised geometry (R(O···O) = 2.901 Å, 0.009 Å from the reference value) [S6].

3.5 2.5 Verification

Two routes to E_NLC are computed and compared at run time, with a hard failure if they disagree: our explicit pair sum, and PySCF's native kernel (`numint.nr_nlc_vxc`). These are independent implementations of the kernel but share a library, so agreement between them validates the arithmetic and not the functional parameters. A second-program anchor now exists for all eight systems (§4), so no quantity in Table 1 rests on agreement between two implementations of the same kernel alone. Energy accounting is checked independently: for a hybrid functional the exact-exchange energy is folded into `vhf.exc` by `get_veff` and is absent from `nr_rks`, so we close e_tot = e_nuc + e_1 + e_coul + e_xc to $\leq 1.1\times10^{-10}$ kcal/mol rather than assuming any particular partition of the XC terms.

Cross-version and cross-machine reproducibility is checked on one system run twice: benzene-methane at grid level 2 with counterpoise, on PySCF 2.12.1 on one machine and PySCF 2.14.0 on another. The two releases generate different numbers of grid points at the same level: the partitioned level-1 grid for the water dimer at its official S22 geometry holds 18336 points on PySCF 2.12.1 and 20248 on 2.14.0, 10.4% more. Two other counts were tried and rejected for this sentence: 18368 is the *unpartitioned* reference grid on 2.12.1 - a different grid of a different kind - and 19936 is the partitioned level-1 grid of the *constructed* geometry, so comparing it with 20248 mixes a geometry change in with the version change and had earlier made this difference look like 1.6%.

The energies themselves are what the comparison is about: on that official geometry the two releases print E_NLC = +52.9345 and E_xc - E_NLC = -11757.5276 kcal/mol identically, with a 5.7e-06 discrepancy between the two implementations of the kernel on each side. That is why the agreement is stated quantity by quantity rather than as a blanket "digit for digit". ΔE_NLC agrees to 3.3e-4 kcal/mol (-1.404667 vs -1.405), ΔE_xc to 4e-4 (-2.324590 vs -2.325), ΔE_int to 4.5e-5 (-1.780045 vs -1.780), the attribution span to 2e-4 (0.213163 vs 0.213) and P_frz[becke] to 1e-3 (-1.441416 vs -1.441), each inside the 0.001 kcal/mol to which the second run prints. The one quantity that does not agree is the counterpoise interaction energy itself: -1.341 vs -1.202 kcal/mol, a difference of 0.139 that is two thirds of the level-2 attribution span of the same complex. We report the split, not a diagnosis of it - the candidate mechanism is that ΔE_int is a difference of SCF total energies, which inherit the grid, whereas an attributed pair term need not, and that is exactly what would have to be tested before the claim is made.

What the observation does settle is a constraint on usage: the attributed quantities are the version-stable ones, and a counterpoise energy is not a safe denominator for them [S8].

4 3. Results and discussion

4.1 3.1 Two different things previously reported as one number

Comparing our explicit pair sum against PySCF's native kernel produces a discrepancy whose size depends entirely on *which grid* the comparison is made on, and the two cases must not be quoted side by side. Holding the density and the grid fixed and varying only the evaluation of the kernel, the difference is +1.5e-8, +1.0e-8 and +8.5e-9 Ha at grid levels 0, 1 and 2 — flat in level, because it is a discretisation property of the interpolated kernel itself. Moving the same sum onto the Becke-partitioned grid of `gen_grid.get_partition` instead gives -0.0282, +0.0015 and +0.0001 kcal/mol, which tracks the ability of the grid to represent the density at all: the same three levels integrate the electron number to -1.6e-2, +3.0e-4 and -6.6e-6 electrons away from the exact value. The second row is a property of the grid, not of our algorithm, and it disappears as the grid is

refined [S9]. A residual of the size of the first row is what a reference implementation should look like; a residual of the size of the second row is what a coarse grid looks like from the inside.

4.2 3.2 The nonlocal term is not a fixed fraction of the exchange-correlation term

Table 1 reports omegaB97X-V/6-31G* results for eight S22 systems at the official GMTKN55 geometries. The quantity of interest is dE_NLC, the change in the nonlocal correlation energy on bringing two fragments together from infinity at fixed density, and its ratio to the total dE_xc for the same process. The remainder, dE_xc^rest = dE_xc - dE_NLC, is the semi-local exchange-correlation plus the fraction of exact exchange; it is tabulated explicitly in `MAIN-skeleton.md` table 4b and is recoverable from these two columns by subtraction.

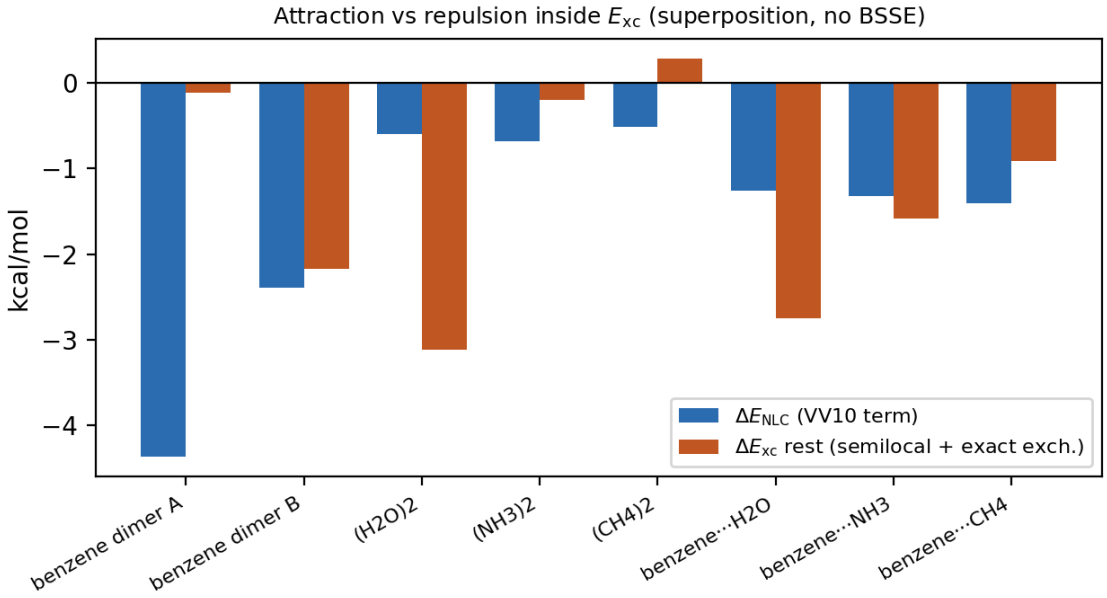


Figure 1: **Figure 1.** The same quantity two ways, per system: the VV10 term ΔE_NLC and the remainder ΔE_xc - ΔE_NLC (semilocal exchange-correlation plus the exact-exchange fraction), both for the superposition difference at grid level 1 and uncorrected density. The remainder is negative for seven of the eight systems but positive for the methane dimer (+0.285 against a VV10 term of -0.517 kcal/mol), the one case where the semi-local remainder is itself repulsive and the nonlocal term alone provides the binding. The other seven span -0.119 to -3.115 kcal/mol, so no single fraction of ΔE_xc belongs to the VV10 term. Data: `nlcsplit/logs/step_h.json`, produced by `nlcsplit/step_h_s22.py`; drawn by `nlcsplit/figures.py` [S2].

All entries in kcal/mol, all at grid level 1. The two benzene rows were tabulated at level 0 in an earlier version of this table; moving them to level 1 shifts dE_NLC by 0.081 (A) and 0.013 (B) kcal/mol. Level 0 is therefore not adequate for a 24-atom aromatic dimer, unlike the six-atom water dimer where levels 1 and 2 agree to the last digit (section 3.2).

Why the methane row has no share. Its ratio is |dE_NLC|/|dE_xc| = 2.2, which trips the cancellation test that ships with the code (`nlcsplit/step_f_dispersion.py`: above 2, the denominator is reported as two signed terms, never as a percentage - here -0.517 plus a semi-local remainder of +0.285). Quoting that ratio as a share - it is 2.2, so more than twice the

System	atoms	dE_int	dE_xc (total)	dE_NLC	share of dE_xc (only where it is defined)
benzene dimer (A, parallel-displaced)	24	-3.580	-4.486	-4.367	97%
benzene dimer (B, T-shaped)	24	-3.188	-4.555	-2.388	52%
benzene...water	15	-4.342	-4.012	-1.259	31%
benzene...ammonia	16	-2.928	-2.915	-1.327	46%
benzene...methane	17	-1.785	-2.322	-1.404	60%
water dimer	6	-7.816	-3.708	-0.593	16%
ammonia dimer	8	-5.167	-0.881	-0.678	77%
methane dimer	10	-0.664	-0.232	-0.517	n/a (see note)

whole change - would have advertised a figure whose divisor is an accident of two larger numbers. What that row does support, without any forbidden ratio, is a comparison against a physical quantity: at level 2 with a counterpoise correction the whole interaction energy is -0.466 while the nonlocal term alone is -0.517 [`evidence/l2cp/20260928T213839Z_centos-step_h_l2cp.out`], i.e. the VV10 term exceeds the net binding by 11% of it. That excess is 0.051 kcal/mol, which is 51x the cross-program residual of the nonlocal term for this very system (0.0010 kcal/mol, `evidence/orca_widen/cross_program_table.txt`) and about 10^3 the cross-version reproducibility of the counterpoise energy (4.5e-5 kcal/mol, §2) - so the excess is a measurement rather than noise, even though the ratio of the two numbers behind it is still not something we may print as a share. The final column is the reason this decomposition is worth having: the nonlocal term ranges from one sixth to more than the whole of the rest of XC, and in the methane dimer it exceeds it, so that the remainder of the XC contribution is *repulsive* while the nonlocal term alone provides the binding. No single fraction of dE_xc can be assigned to VV10 across a benchmark set, and the two geometries of the same 24-atom benzene dimer (97% vs 52%) show that the ratio is a property of the arrangement, not of the pair of molecules.

For the six systems with at most 17 atoms we additionally re-ran at grid level 2 with a full counterpoise correction - six of the eight; only the two benzene-dimer geometries stay at level 1, and benzene...methane needed a separate run after its first attempt was cut off by a four-hour wall-clock budget [S10]. Between levels 1 and 2 neither dE_NLC nor the interfragment pair term moves by more than 1.0e-3 kcal/mol, but that number is a statement about storage precision, not a measurement. Every level-2 row in the sidecar is kept to three decimals, so no level-1 to level-2 difference below 5e-4 kcal/mol is observable from these files at all. For the water dimer, whose level-1 row carries full precision (-0.5934948 and -0.6012148 kcal/mol), level 2 prints -0.593 and -0.601 - consistent to the printed digit, and nothing sharper can be claimed from this evidence. For the other five systems both rows are three-decimal, so 1.0e-3 there is the rounding floor rather than a measured change. The counterpoise-corrected interaction energy comes out at -6.129 kcal/mol for the water dimer, i.e. a basis-set superposition error of 1.687, 1.491, 0.196, 1.075, 0.760 and 0.578 kcal/mol for water, ammonia, methane, benzene...water, benzene...ammonia and benzene...methane respectively [S10]. The core quantities of this method are therefore converged one grid level before the total binding energy is; a user who only wants dE_NLC can stop at level 1.

Is -6.129 the right answer? Not a question about the nonlocal term, but it is a question the paper invited, and it now has a measurement instead of a caveat. At the same official S22 geometry, the same 6-31g basis and the same counterpoise convention, coupled cluster gives dE_int = -6.856 kcal/mol at CCSD(T) with frozen oxygen 1s (-6.887 all-electron, so the frozen-core choice is worth 0.031 here), and -7.085, -5.248, -5.932 and -5.203 at cc-pVDZ, aug-cc-pVDZ, cc-pVTZ and aug-cc-pVTZ respectively [S24]. Two things follow, and they are not in conflict. At fixed basis

our functional binds 0.73 kcal/mol *less* strongly than correlation does in the same basis. As the correlated calculation approaches the basis limit it becomes -5.20, at which point the density functional result is 0.93 kcal/mol too deep - and that near-limit value is externally consistent, because an open-access coupled-cluster study quotes the equilibrium-geometry CCSD(T)/CBS water-dimer dissociation energy as -5.01 kcal/mol [20]. The correlated ladder moves 1.9 kcal/mol across five bases while the functional's deviation from it is about 1. So a verdict of the form "the functional overbinds" obtained by comparing one small-basis density-functional number with one CBS reference is not supportable: it folds a basis-set effect twice the size of the quantity being judged into the judgement. That is a caution about the total binding energy, not about dE_NLC, which is what every claim in this paper is made of, and which is grid-converged a level earlier (above) and agrees with a second program to 0.0004 kcal/mol on this same complex [S18]. A related definitional point, learned from a retraction: the S22 tables report *total* interaction energies [19], and the revision that made them basis-set-consistent keeps doing exactly that [21], so no "CCSD(T) dispersion energy" can be read out of them, and an earlier version of this draft quoted one.

The same anchor on ammonia and methane is what makes it usable. The single-system version above would read as "the functional is 0.7 kcal/mol too shallow", and that reading does not survive being repeated. At 6-31g, with every other setting identical, the three systems give correlated values of -6.856, -4.455 and -0.178 kcal/mol against our function's -6.129, -3.675 and -0.466 [S24]: shallower by 0.73 and 0.78 on the two hydrogen-bonded systems, *deeper* by 0.29 on the methane dimer, which at that point looks like the sign of the discrepancy is system-dependent. It is not, and the way it fails is worth recording because an earlier draft of this paragraph said the opposite. Re-running the correlated side in aug-cc-pVDZ moves the methane value by a factor of five (-0.178 to -0.947) and inverts that comparison - the methane dimer is dispersion-bound and 6-31g has no diffuse functions, so its small-basis number was a basis artefact and the companion archive had flagged it as such before the second basis was run. What the three systems establish is therefore narrower and firmer than a sign: at a fixed basis the deviation is 0.3-0.8 kcal/mol with no single sign, and the correlated reference itself moves 0.18 to 0.95 (methane), 4.46 to 3.33 (ammonia) and 6.86 to 5.25 (water) when only the basis changes. Two effects of the same size cannot separate each other, which is why this section judges the functional nowhere and judges dE_NLC instead. The correlated numbers are also not a one-program artefact: across three systems, two bases and both frozen-core conventions, the same counterpoise energies recomputed in ORCA 6.0.1 agree with PySCF's to 0.0002 kcal/mol, and ORCA's printed AO dimension equals the PySCF nao for every one of the 24 single points, so the two numbers being compared come from the same one-particle basis rather than from two bases that happen to give close energies - which is the distinction that matters here, because a d-shell convention difference is worth 0.01-0.1 kcal/mol, one order of magnitude above the largest disagreement measured [S24].

4.3 3.3 The interfragment pair term saturates the frozen-density limit

`inter[(A,B)]`, summed over the retained grid points, is directly comparable to the superposition difference d_sup computed from the unrelaxed monomer densities, because both hold the density fixed. Their ratio falls in 0.92-1.08 across the six systems that carry a counterpoise reference at level 2 [S8]. The two benzene-dimer geometries are not in that set: they were run at level 1, and the ratio printed for them there has a different denominator. On the coarser level-0 grid the same ratio is 1.017 for the water dimer at the official S22 geometry and 1.025 at the omegaB97X-V-optimised one, while the benzene sandwich falls below the band at 0.915 [S19] - the level-2 value of 1.01 for the official water geometry sitting between those two [S8]. The ratio being able to exceed unity matters: relaxation is a small correction of *either* sign on top of the frozen-density interfragment term, not an

orthogonal contribution that adds to it. Physically this says that in these complexes essentially all of the nonlocal correlation that changes on association is carried by grid-point pairs straddling the two fragments, with the intra-fragment nonlocal correlation of each relaxed monomer contributing the balance [S2, S10].

This is where the explicit sum earns its cost. A point-value kernel evaluation returns dE_NLC and nothing else; the (i, j) index structure retained by the double sum is what makes 0.92-1.08 a statement that can be checked at all. The corresponding dispersion component published for the benzene dimer is -3.34 to -3.80 kcal/mol at the PBE0-VV10, revPBE-D3 and related levels of an ALMO-EDA study, and -3.69 at omegaB97M-V [S11]; our -4.367 (Table 1, level 1; the level-0 value was -4.448) is the same kind of quantity to within functional-generation and basis-set differences, and it is *not* comparable with our own uncorrected interaction energy for that dimer, -3.580 at the same level, which is smaller because Pauli repulsion cancels most of it.

4.4 3.4 Attribution-scheme sensitivity is system-dependent

Repeating dE_NLC for the four partition schemes of section 2.3 gives absolute spans of 0.004-0.247 kcal/mol and relative spans - measured against the Becke-scheme interfragment term - of 0.8-18.2%. Neither ordering follows system size: the 24-atom parallel-displaced benzene dimer has one of the smallest absolute spans (0.032, below the 6-atom water dimer at 0.043), while the 15-atom benzene...water gives the largest (0.247). What separates them is the arrangement, not the atom count or the interaction class:

System	**P_frz (Becke)**	**span (4 schemes)**	**relative**
methane dimer	-0.474	0.004	0.8%
benzene dimer (A)	-4.038	0.032	0.8%
ammonia dimer	-0.649	0.029	4.5%
water dimer	-0.601	0.043	7.2%
benzene dimer (B)	-2.371	0.219	9.2%
benzene...methane	-1.442	0.213	14.8%
benzene...ammonia	-1.393	0.224	16.1%
benzene...water	-1.360	0.247	18.2%

Two observations follow. First, the relative span has no pattern by interaction type: hydrogen bonding spans 7.2% (water) and 4.5% (ammonia), dispersion-driven complexes span 0.8% (methane) and 18.2% (benzene...water), and the two benzene dimer geometries differ by a factor of eleven. A single blanket uncertainty percentage for this class of method would misstate the methane dimer by more than twenty-fold. Error bars must be reported per system (Fig.~2). Second, the span is an attribution property rather than a one-particle-basis property. On the official S22 water-dimer geometry - the same geometry the rest of this section's tables are computed on - re-running the four schemes at level 1 across five basis sets gives spans of 0.0299 (STO-3G), 0.0435 (6-31G $\times$), 0.0413 (aug-cc-pVDZ), 0.0421 (def2-SVP) and 0.0423 (def2-TZVP) kcal/mol while P_frz itself moves from -0.458 (STO-3G) to -0.633 (aug-cc-pVDZ) [S12]: the quantity of interest drifts by 0.18 kcal/mol across the basis ladder while the scheme spread stays within $\pm$3% of 0.042 for the four basis sets beyond the minimal one, at 0.030 in STO-3G. The last two rungs move P_frz by 0.026 kcal/mol, below the 0.05 the ladder's own criterion asks for, so the basis is near its limit while the span is not shrinking - the drift and the spread are different phenomena. The same ladder on the constructed geometry (0.0249/0.0400/0.0382/0.0390/0.0398, drift -0.386 to -0.554) reproduces this

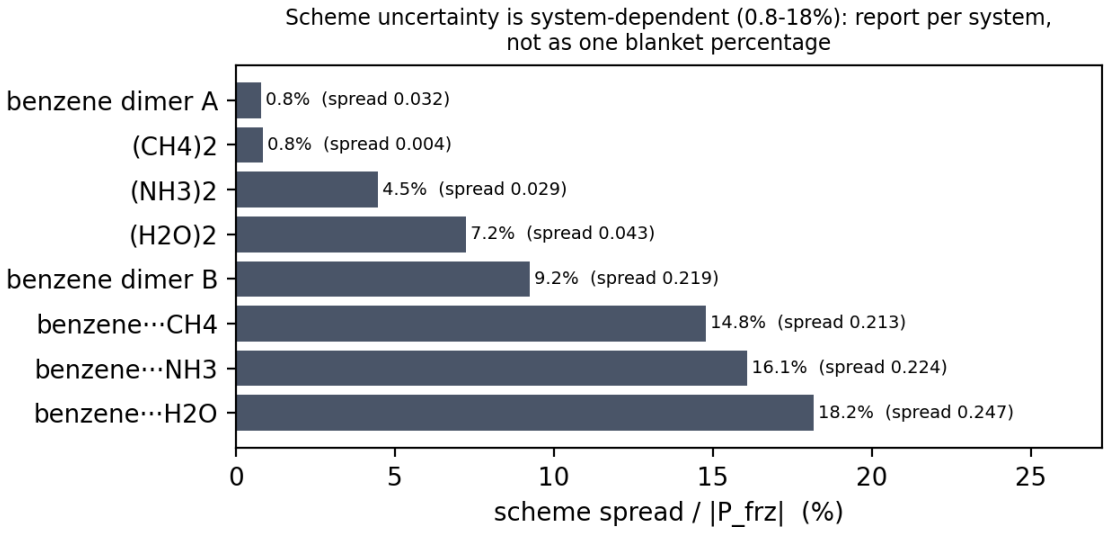


Figure 2: **Figure 2.** How much of the reported pair term is a choice about where grid points belong: the spread of P_frz over the four partition schemes of section 2.3, divided by the Becke-scheme value, per system, sorted. The two geometries of the same 24-atom benzene dimer sit at 0.8% and 9.2% and benzene...water at 18.2%, so the percentage is a property of the arrangement rather than of the pair of molecules and cannot be collapsed into one blanket error bar. Data as in Figure 1 [S2].

to 5-8%, so the reading is not an artefact of which water dimer we sat down with. No better basis set will remove the spread, which is why it - and not grid convergence, which is 0.0081 kcal/mol over level 0 to 2 at 6-31G and 0.0002 over level 1 to 2 [S13] - is the error bar this method has to carry.

4.5 3.5 A limit of fragment-resolved quantities

Because beta is a constant and the integral of rho is the electron number, the local term attributed to a fragment is exactly beta*N_A where N_A is its Becke population, and beta = 3.0415 kcal/mol per electron. Any scheme-induced spread in the population is therefore amplified into the fragment-resolved local term by that factor as an identity, not an estimate: for the water dimer at grid level 3 the four schemes give fragment-A populations of 10.1559 (Becke), 10.1537 (Stratmann), 10.0979 (Becke–Treutler) and 10.0478 (Becke–Rad) electrons on the official S22 geometry, a spread of 0.108 electrons which is 0.329 kcal/mol; on the constructed geometry the same spread is 0.0987 electrons, i.e. 0.300 kcal/mol [S20]. Two consequences deserve stating explicitly. In a superposition difference the term cancels identically — electron number is conserved — so every dE_NLC value in this paper is free of it. But any table reporting *absolute* fragment energies inherits an uncertainty of this size from a quantity (atomic population) that the method does not control. This is a property of Becke partitioning that every EDA scheme shares, and we state it rather than claim it as a contribution.

A related implementation note: attributing the local term softly, by evaluating all fragment factors s_A at every grid point rather than assigning each point to one owner, changes these numbers by at most 1.53e-6 electrons (4.6e-6 kcal/mol after multiplication by beta) on the official geometry the tables use, and by 8.69e-7 electrons (2.64e-6 kcal/mol) on the constructed one - grid

level 3, all four partition schemes, both numbers now in [S23]. The soft and hard readings of the local term are numerically the same thing, which is worth recording because the two are sometimes discussed as if they competed. The distinction between the two geometries is itself the lesson here: this sentence read "8.7e-7 electrons" for a week, which was true of the constructed geometry and silently too small for the one we publish.

5 4. Scope, independent verification, and limitations

This section keeps four things apart that are easy to read past one another: what the reported quantity is not (§4.1), the two checks that were run against something outside our own library (§4.2 and §4.3), the instrument behind the novelty claim (§4.4), and the limits that survive all of it (§4.5).

5.1 4.1 What this decomposition is not

This is not an energy decomposition analysis. We split dE_xc into the nonlocal term and everything else, and we resolve the nonlocal term into fragment pairs. We do not separate electrostatics, exchange-repulsion, polarisation or charge transfer, and the resulting components are therefore not comparable term-by-term with ALMO-EDA or SAPT(DFT) output. Counterpoise-corrected interaction energies are reported in section 3.2 for six of the eight systems, but dE_NLC itself needs no such correction: it is evaluated at fixed (non-self-consistent) superposed density, so basis-set flexisation does not enter it, which is precisely why it converges a grid level earlier than dE_int does [S10].

The nonlocal term is not the dispersion energy. Section 3.2 is the evidence: dE_NLC accounts for anywhere between 16% and 97% of dE_xc across the seven complexes where that share is defined by our own cancellation test (Fig.~1) - and note that this is not a smooth ladder: the two geometries of the same benzene dimer give 97% and 52%. The eighth complex, the methane dimer, is the case a percentage cannot describe at all: its two parts carry opposite signs, so the nonlocal term is not a fraction of the change but the thing that has to overcome it. A quantity whose relation to another varies that far - and in one case changes character - cannot be identified with it. We use "dispersion" only when quoting external work, and the values in Table 1 should be read as "the contribution of the VV10 term", not as a dispersion energy.

5.2 4.2 Cross-program check: ORCA 6.0.1 agrees on the reported quantity, on all eight systems

Our explicit sum had only ever been checked against PySCF's native kernel, and both live in one library, so they could share an error in the functional parameters or in the kernel convention. That check now exists against a second program, and on more than the two systems we first reported. ORCA 6.0.1 prints the term natively for `! WB97X-V` as `NL Energy, E(C,NL)`; the control showing that the label is VV10-specific rather than a generic DFT printout is `evidence/orca_control/h2o_wb97x.out`, an ωB97X run with no `-V`, which contains no such line [S18]. Twenty-four serial runs - the water dimer and benzene sandwich (S22 #2, #11) on the cluster, the other six systems on this machine, all at S22 official complex geometries with 6-31G*, each terminating normally with a converged SCF and carrying exactly one `NL Energy` line - give, for the superposition difference that Table 1 reports. The last two columns compare the residual with the partition-scheme span of the same row (§3.4), which is the quantity a referee has to be smaller than to be informative:

system	ΔE_NLC ORCA	ΔE_NLC PySCF	difference	span	residua
water dimer (S22 #2)	-0.5931	-0.5935	0.0004 (0.07%)	0.0435	0.009
ammonia dimer (S22 #1)	-0.6745	-0.6780	0.0035 (0.51%)	0.0290	0.120
methane dimer (S22 #8)	-0.5160	-0.5170	0.0010 (0.19%)	0.0040	0.242
benzene...water (S22 #17)	-1.2522	-1.2590	0.0068 (0.54%)	0.2470	0.028
benzene...ammonia (S22 #18)	-1.3189	-1.3270	0.0081 (0.61%)	0.2240	0.036
benzene...methane (S22 #10)	-1.3912	-1.4040	0.0128 (0.91%)	0.2130	0.060
benzene dimer B (S22 #20)	-2.3685	-2.3880	0.0195 (0.82%)	0.2189	0.089
benzene dimer A (S22 #11)	-4.3252	-4.3668	0.0416 (0.95%)	0.0320	**1.301**

All eight agree to better than 1% of the term, and the residual tracks the magnitude of ΔE_NLC rather than the size of the system, which is what a quadrature difference looks like. Three things must be said about it rather than glossed. First, the grids are not matched - ORCA uses its own nonlocal quadrature (Lebedev-110, IntAcc 4.004) against PySCF's `grids.level=1` - so the residual is the joint grid-plus-implementation difference and we do not attribute it to either side. Second, in seven of the eight systems the cross-program residual is smaller than the attribution span it would have to resolve, by 8 to 100 times; the exception is the benzene sandwich, where 0.042 kcal/mol is 1.3x the 0.032 kcal/mol span for that same geometry. That single row is the warning this comparison exists to provide: cross-program noise can be of the same order as the quantity being resolved, so it must never be folded into the attribution uncertainty. Third, the two water monomers of S22 #2 are not equivalent (donor r_OH 0.9639/0.9571 vs acceptor 0.9590/0.9590), so the monomer term had to be computed twice - using 2x one monomer would have introduced ~0.0004 kcal/mol, exactly the size of the residual. A repeat of one input inside the same allocation reproduced its VV10 term to 1.6e-11 Eh, so the residual is not run-to-run noise [S18]. The widened set was assembled after an audit of our own record found that `evidence/orca_widen/orca_widen.json` marked all twelve of its first runs `converged: false` while the archived outputs print `SCF CONVERGED AFTER n CYCLES` and `ORCA TERMINATED NORMALLY`: the flag came from a criterion that looked for a different program's wording. The flags were re-derived from the logs without re-running anything, with the assertion that no energy value moves (`evidence/orca_widen/generator_rescan_convergence.py`), and the table above is generated from the tracked artefacts by `evidence/orca_widen/generator_cross_program_table.py`, which refuses to emit it if the system list disagrees with the one it declares, or if any row's basis, grid level or counterpoise setting disagrees. One operational note belongs with the row it affects: the 24-atom benzene dimer (B, S22 #20 - the T-shaped one, not the parallel-displaced (A)) is the only run that needed four times the per-process memory allowance (%maxcore 4000 against 1000), which changes what ORCA can hold in core and nothing about the numerical parameters; at 1000 it died in the SCF with `CANNOT OPEN FILE dimer.opot2.tmp`, and that difference is recorded on the run rather than left for a reader to rediscover.

What this also checks is the absolute value, not just the difference. An earlier log of ours recorded a ~0.14 kcal/mol disagreement on the water monomer between our explicit sum and the native kernel, and we flagged the absolute fragment energies as unrefereed. That flag is now retired: the archived level-2 run gives a water-dimer nonlocal term of +52.9345 kcal/mol against ORCA's +0.084358235 Eh = +52.9356, and monomers +26.7635/+26.7644 against ORCA's +26.7646/+26.7641 - agreement at the 1e-3 kcal/mol level on the absolute quantity as well [S8, S18]. We have not re-established what that 0.14 was: the most likely cause is a comparison made before the beta·∫rho counterterm was added back - `step_f_dispersion.terms` now adds it before

reporting, and the comment at that call site exists precisely because of this confusion - but we are stating the present agreement, not a diagnosis of the old number, so the section 3.5 identity is covered by this anchor without needing that story to be true.

5.3 4.3 A referee for attribution, which no other program can supply

A second program checks the value of the term; it cannot check how the term is *owned*, because it does not attribute at all. So `evidence/attribution_referee/` moves to a system where an exact, partition-free reference exists by construction: two normalised Gaussians, four electrons each, centred at `+/- sep/2` on the z axis, with `Phi` built once from the total density and `I_XY = int int rho_X Phi rho_Y` taken for each pair of density factors. No Becke weight, no grid assignment and no fragment ownership appear anywhere in that definition, and `E_NL = 1/2 (I_AA + I_BB) + I_AB` holds exactly, so `I_AB` is what the inter-fragment block would have to be if attribution were a property of the density rather than of a convention. It is computed three ways - a 3-D x 3-D Cartesian tensor product, a 5-D cylindrical reduction about the internuclear axis (different nodes, domain shape and Jacobian), and importance-sampled Monte-Carlo using the density factor itself as the importance - and the Monte-Carlo agrees with the quadrature within its own standard error everywhere, worst `|z| = 2.23`.

The outcome separates cleanly, and the two halves say different things. Handed the exact density factors, the package's own summation code reproduces `I_AB` to `7.96e-07` Ha worst, inside the reference's combined uncertainty `U = 1.060e-05` Ha (0.00665 kcal/mol, assembled from three measured error bars: order-ladder last change, domain-truncation shift, and the Cartesian-versus-cylindrical disagreement). The arithmetic is therefore refereed, and by a route that does not share our library. Handed the production Becke/owner assignment, the same code lands `1.68e-08` to `1.26e-04` Ha from `I_AB` (0.00001-0.0788 kcal/mol), and because the same comparison is also run on a second, disjoint node set where the grid cancels, that offset tracks the partition rule and not the grid: the grid's own cost in the same quantity stays below `U` at every configuration. Reported as sizes, the offset is 0.0007-1.84% of the whole VV10 term and 0.32-11.79% of the cross term it is meant to be.

The resolution is stated rather than assumed, and it depends on which reference is holding the bar. Against the deterministic reference and its combined uncertainty `U`, the offset exceeds `U` in **5 of the 12 configurations** (2x, 3x, 3x, 4x and 12x `U`); in the other 7 it is smaller than that error bar, and there this referee does not certify the convention, it reports that it cannot distinguish it [S22]. Repeating the comparison against the Monte-Carlo route instead - 4,000,000 draws per configuration, scored against `max(2 s.e., |MC - 5-D|)`, whose worst value is 1.819e-06 Ha or 0.00114 kcal/mol, 5.8x tighter than `U` - resolves **12 of 12** [S22]. Both numbers are reported because they answer different questions: the deterministic `U` is bounded by how far the two quadratures disagree with each other, the Monte-Carlo bar by an independent estimator's own scatter plus that disagreement, and neither is traded for the other. Those five resolved offsets run 0.013-0.079 kcal/mol, which is 0.55-1.84% of the whole VV10 term and 2.5-9.7% of the cross term, the same order as the four-scheme spread of section 2.3 (0.004-0.247 kcal/mol) - the distance from one convention to a partition-free reference is about the distance between conventions, which is the consistency this section wanted.

Two limits travel with that result. The densities are synthetic (no nuclei, no cusps, no orbital structure), so what is refereed is the attribution machinery as a mathematical operation on a density, not a chemical statement. The positive control - the documented-wrong pairing of section 3.5 - moves the whole term on these densities by 0.0052 kcal/mol, which is larger than most of the offsets tabulated above: it therefore shows the referee can see a kernel-definition error, but

it also shows that this apparatus resolves attribution conventions, not kernels. Extending it to heteronuclear pairs is not a parameter change: the reference itself needs no nuclei, but the package side is evaluated on a grid whose Becke radii come from atomic numbers, so making the two centres unequal forces a choice of radii - the very convention under test - and that has not been done here. It does reproduce: two independent executions of the whole pipeline give tables that are byte-identical except in the recorded wall-clock fields, and both runs are shipped in that directory.

5.4 4.4 How we know the literature search means anything

The zeros in section 1 are only as good as the instrument, so we calibrated it in the same session: a known paper title was retrieved exactly (1 record), a broad concept query returned 3,317 records, and a topical AND query in the same vocabulary returned 33. By contrast the two open indexes we tried first failed the same test - OpenAlex and Crossref each retrieved a known title 4/4 times when queried by title but missed that same known, on-topic paper for 2 of 3 and 3 of 3 legitimate topical phrasings respectively. We therefore report hit counts only from the instrument that passed. Its enumeration is complete rather than sampled, and that took a second round: the result-page UI virtualises its list and gave us one record per list, and the first attempt at working around it by excluding already-read titles backfired - the title we excluded was the truncated form of the very record we were chasing (FDE-vdW's full title begins "Exact kinetic energy enables..."), so we had removed our own nearest neighbour. Going to the record endpoint instead returned every record of every list. One further trap there is worth naming, because it fails silently: that endpoint streams records in batches of 20, and a parser that overwrites rather than appends reports the tail batch as the whole result set - on a 33-record query that reads as 13, and nothing in the response says so. We now assert `len(records) == RecordsFound` on every query.

The search covers one index - Web of Science Core Collection - because Scopus was measured as unsubscribed at our institution and Embase and SciFinder were not searched at all. Within that one index, though, the search is now a full enumeration rather than a sample: every record returned by every query was read, which changed the conclusion in one place worth reporting, since a keyword-shaped novelty query is narrower than the claim it is meant to test. Six queries in the first round covered the decomposition vocabulary and found nothing in the interpretive line; adding the broadest topical query available (vdW-DF x "nonlocal correlation", 33 records, every title read) surfaced Hyldgaard, Berland and Schröder's xc-hole interpretation of the same energy [S17], which no query containing decompose, partition, fragment or pair would ever have returned. The open indexes tried instead (OpenAlex, Crossref, Semantic Scholar, arXiv) are recorded as supplementary and not as a second formal search, because their reachability was uneven and measured: OpenAlex's search route returned 21 usable responses against 14 rate-limited and 10 server-error attempts, Semantic Scholar's search route was rate-limited throughout so its keyword dimension was never actually queried, arXiv failed known-item calibration (its own abstract field contains a term that its search does not retrieve) so its zeros are not read as absence, and Scopus returned 401/403. Crossref and OpenAlex retrieve known titles reliably but miss a known on-topic paper when asked topically. The round still changed this paper: the nearest neighbour, and the reason the claim is narrower than it was written, came from it rather than from the formal search.

5.5 4.5 The limits that remain

The attribution uncertainty is intrinsic and per-system. The 0.004-0.247 kcal/mol spread of section 3.4 does not shrink with a finer grid or a larger basis; it is a property of how grid points

are assigned to atoms. Any number produced by this method carries it, and reporting a single percentage for the whole method would be wrong by more than twenty-fold for individual systems.

Absolute fragment energies inherit a population uncertainty. As shown in section 3.5, the local term of a fragment is exactly beta*N_A, so the 0.108-electron spread of the attributed population over the four schemes is a 0.329 kcal/mol spread in the reported fragment energy - larger than the scheme spread of the pair term itself for the water dimer. Differences are safe; absolutes are not, unless the population convention is stated alongside them.

Cost, and what it buys. The explicit double sum is quadratic in the number of retained grid points; we have not implemented a sub-quadratic approximation, and the point-value kernel that PySCF ships is both cheaper and accurate to ~1e-8 Ha on its own grid (section 3.1). The cost is not prohibitive in itself - JuNoLo ran the same N^2 sum over massively parallel hardware for surface calculations [S14] - but it is a real constraint, and it is worth naming the trade honestly. We are therefore not claiming that this method computes dE_NLC better. What it buys is the (i, j) index structure: the 0.92-1.08 frozen-density result of section 3.3, and the per-pair numbers behind Table 1, are not recoverable from a scheme that collapses the pair sum to point values first.

Scope of functionals. Everything here is the VV10 kernel with the (b, C) parameters PySCF reports for omegaB97X-V, using the same-centre pairing convention of section 2.1. Functionals whose nonlocal term uses the regularised rVV10 form have not been tested and may not share the pairing convention.

6 5. Code availability

The package `nlcsplit` (Apache-2.0, containing no GAMESS source) implements the kernel, the four attribution schemes, the fragment decomposition and every table in this paper; it is built on PySCF [22] [23] and NumPy [24], and each table is produced by exactly one `step_*.py` script whose stdout is archived under a UTC-stamped filename. Fragment boundaries come from covalent connectivity rather than hand-written atom lists, and both the geometry constructors and the official-geometry loader assert internal coordinates at construction time - three of the bugs this measure was added to catch are described in the supplementary notes. Grid, basis and scheme sensitivity are separate functions (`step2_convergence.py`, `step_g_basis.py`, `step_h_s22.py`) rather than one combined uncertainty, and `partition_factors()` verifies its own output against PySCF's partition weights before returning it.

7 6. Conclusions

We have shown that the VV10 nonlocal-correlation energy can be resolved into fragment pairs without giving up the total: the double sum is evaluated explicitly, the local part cancels identically from any superposition difference because it is exactly beta times the electron number, and the resulting blocks reproduce the total to machine precision while carrying an (i, j) index that the point-value evaluation discards.

Three things follow from applying it to eight S22 complexes at official geometries. The VV10 term is not a fixed share of the exchange-correlation contribution - 16% for the water dimer, 97% and 52% for two geometries of the same benzene dimer, and no quotable share at all for the methane dimer, where the two parts carry opposite signs and the denominator is their small residue (the package refuses to report one there by rule, §3.2) - so reporting it as "the dispersion energy" is unsafe even within one benchmark set. The association-time change is almost entirely inter-fragment: the frozen-density pair term accounts for 0.92-1.08 of the superposition difference in the

six systems checked against a counterpoise reference, with relaxation a small correction of either sign rather than an additive channel. And the dominant uncertainty is the attribution scheme, not the numerics: 0.004-0.247 kcal/mol absolute, 0.8-18.2% relative, stable to within ±3% across the four basis sets beyond STO-3G and to within 0.001 kcal/mol between grid levels 1 and 2, with no pattern by interaction type that would let a single percentage be quoted.

What the method does not do is equally important, and we would rather the reader learn it here than in a review. It is not an EDA: it does not separate electrostatics, exchange-repulsion, polarisation or charge transfer. It is not faster than the production kernels it could be compared against - it is quadratic in the grid. Its arithmetic was for most of this project verified only against a second implementation of the same kernel inside the same library; that gap is now closed for the quantity Table 1 reports, by twenty-four ORCA 6.0.1 single points covering all eight systems, every residual below 1% of the term (0.0004 kcal/mol on the water dimer up to 0.0416 on the benzene sandwich). An earlier version of this draft claimed such an anchor and had to withdraw it; what actually failed was not the check but our ability to find its number, because we searched for the words "nonlocal" and "VV10" in an output file that labels the quantity `NL Energy, E(C,NL)`. One scoping limit remains: the absolute value, not only the difference, is checked on the water dimer and on both of its monomers (§4, where the earlier "unrefereed" flag is retired); on the other seven systems the second program checks the difference, which is what every claim in this paper is made of, and not each absolute fragment value. Attribution has a referee of its own now (section 4), on a two-Gaussian density where a partition-free value exists by construction: the summation code reproduces that value to within the reference's own error bar, while the Becke/owner convention sits 1.7e-08 to 1.26e-04 Ha (0.00001-0.0788 kcal/mol) from it - a difference resolved in 5 of its 12 configurations against the conservative deterministic error bar and in all 12 against an independent Monte-Carlo bar 5.8x tighter. The case that draft excluded, unequal exponents and unequal electron numbers on the two centres, is now run too [S22]: the exact reference needs no nuclei and the density-factor partition needs no radii, so the only convention-dependent input is the element label, and swapping the label at a fixed density moves the offset by at most 0.24% of itself. What the heteronuclear case actually changes is the size of the effect - the owner/Becke cross term sits 6.2-35.9% of the exact cross term there, against 2.5-5.2% on the homonuclear scan, every one of them resolved against its own error bar. The homonuclear table understated the non-uniqueness, and it understated it in the regime where an EDA is asked to work. What still does not carry a referee is a chemically realistic density: nuclei, cusps and orbital structure remain outside this check, which is about the attribution rule as a mathematical operation.

Two further items are open and stated as such rather than smoothed over. The first is a number, not a missing run: re-running benzene-methane at level 2 with counterpoise on two PySCF releases reproduced every attributed quantity to within 5e-4 kcal/mol - the rounding half-step of the three-decimal printout on one side, the largest measured difference being 4.2e-4 on the pair term - but reproduced the counterpoise interaction energy only to 0.139 kcal/mol (section 2), so a counterpoise denominator is not portable between versions while the pair term appears to be. The second is a scope limit that has since been closed, and how it was closed matters more than the number: the basis-set ladder of section 3.4 had been run on the constructed water-dimer geometry while every other table in this paper is computed on the official S22 one. Re-running the five-basis, four-scheme ladder on the official geometry (level 1, same script, spans 0.0299/0.0435/0.0413/0.0421/0.0423 kcal/mol against 0.0249/0.0400/0.0382/0.0390/0.0398 on the constructed one) leaves every reading of the argument intact to 5-8%, so section 3.4 now quotes the official geometry. An earlier version of this paragraph claimed instead that the ladder predated the 2026-09-29 geometry fix and "has not been re-run" - which was not true: the fix landed at 01:07 that morning (+08) and the re-run is stamped 08:31, and the provenance row for that table already said it was on the constructed

geometry. Declaring a task open is a claim too, and it was checked against the artefacts only after the fact.

The package is released under Apache-2.0 with no GAMESS source in it, every number in this paper is produced by a named script whose archived stdout is part of the repository, and the two structural guards we added after finding bugs in our own geometry builders - coordinate assertions at construction time, and `partition_factors()` verifying its own output against PySCF's partition weights before returning - are the part of this work we would most recommend copying.

8 Appendix: provenance of every quoted number

Each tag used above resolves to a file in this repository. "line N" means the value can be read literally at that line of the archived standard output; those stdout files live under `evidence/**`, are tracked, and are hash-listed under `evidence/*/MANIFEST.sha256` — they were not re-typed by hand. `nlcsplit/logs/` is a different case and is stated as such rather than glossed: only three sidecar JSONs there are tracked (`step_h.json`, `step_h_l2cp.json`, `orca_anchor.json`), the rest of that directory ignores `*.out` by its own `.gitignore`, so a run log left there is a working note and not evidence.

`evidence/attribution_referee/mc_tighten.py` (4,000,000 draws per configuration, its own seed base) and is stored in `evidence/attribution_referee/outputs/mc_tight.json`, a separate sidecar, so neither archived run is disturbed by it. Re-running `evidence/attribution_referee/run.py` regenerates every number; `evidence/attribution_referee/summariz` re-quotes them without re-quadrating. `hetero.py` is the heteronuclear extension, and it is written as a sidecar for the same reason `mc_tighten.py` is: it must not disturb the archived table. Row 0 is therefore homonuclear and labelled as the archived run labels it, and it reproduces the reference cross term and the package cross term with a difference of exactly 0.000e+00 - which is what licenses the three unequal-centre rows. It skips the referee's own disjoint half-box node set, because that construction assumes two congruent boxes, and it runs each density under two different element labels so the radii table's contribution is measured instead of assumed | | S23 | `evidence/population/20260930T200927Z_stepk_soft-vs-hard-official.out`, `evidence/population/20260930T201142Z_stepk_soft-vs-hard-constructed.out`, `nlcsplit/step_k_population.py`, `evidence/population/MANIFEST.sha256` | the soft-vs-hard reading of the local term, per partition scheme, on both water-dimer geometries at grid level 3: hard populations credit a point to its owner only, soft populations spread it over the fragment factors at that point, and both are computed on PySCF's own partition weights so the measure is the converged one. The two files give the largest soft-hard population difference (1.527e-06 electrons official, 8.686e-07 constructed) and its conversion through beta into kcal/mol. Before these runs existed, this sentence quoted 8.7e-7 electrons on the strength of a working log only - the number was real but its geometry was not the published one, which is why the scope is now stated in the sentence | | S24 | `nlcsplit/step_m_ccsdt.py`, `evidence/ccsdt/20260930T224400Z_stepm_ladder_after_refactor.out`, `evidence/ccsdt/20260930T235900Z` `evidence/ccsdt/20261001T000400Z_stepm_three_systems_avdz.out`, `evidence/ccsdt/20260930T212300Z_st` `evidence/ccsdt/20261001T044833Z_orca_crosscheck.out`, `evidence/ccsdt/20261001T052943Z_orca_crosso` `evidence/ccsdt/orca_crosscheck.py`, `evidence/ccsdt/README.md`, `evidence/ccsdt/MANIFEST.sha256`, `nlcsplit/tests/test_cc_convention.py` | the coupled-cluster anchor for the water-dimer binding energy: same official S22 geometry, same basis, same counterpoise convention, RHF reference, MP2/CCSD/CCSD(T) from PySCF's own modules, each rung reported both frozen-core and all-electron. The archive also records the mixed-convention bug this run nearly shipped with:

Tag	Artefact	What it fixes
S1	`RESEARCH-20260929-lit.md` §1, §6	first-round WoS/Scopus search record: strings, hit counts, search date
S2	`MAIN-skeleton.md` tables 2 and 4b; `nlcsplit/logs/step_h.json`	the level-1 S22-subset table, incl. the 0.8% / 18.2% nonlocal fractions
S3	`nlcsplit/step_f_dispersion.py`	the level-1 dispersion-scale check that leaves the double sum as the only unaccounted term
S4	`MAIN-skeleton.md` table 4b	the 52%-97% range of the same quantity across the two archived geometries of one molecule pair (benzene dimer A parallel-displaced, B T-shaped) — the geometry axis, not the partition axis
S5	`nlcsplit/step_e_kernel.py`	explicit pair sum vs PySCF's native VV10 kernel on a fixed grid
S6	`evidence/centos/20260928T165857Z_centos-step_c.out` line 19	relaxed O-O = 2.901 A
S7	`nlcsplit/step_d_floor.py`, `evidence/recompute/20260928T150222Z_step_d_floor.out`, `nlcsplit/tests/test_kernel.py`	the local term is beta x electron number, so it cancels in any superposition difference; the archived stdout is the rho_floor sweep itself - dropped-point fractions, electrons carried by the dropped points, and the per-level Ha spans behind the sensitivity claim in §3
S8	`nlcsplit/logs/step_h.json`, `nlcsplit/logs/step_h_l2cp.json`, `evidence/l2cp/20260929T051912Z_centos-l2_benzmeth_cp.out`, `evidence/l2cp/20260929T051849Z_l2_benzmeth-wsl.out`, `evidence/centos/20260928T165857Z_centos-step_f_h2o2.out`, `evidence/recompute/20260929T010500Z_step1_geomfix.out`	cross-version pair (PySCF 2.12.1 vs 2.14.0) and the 19936 vs 20248 partitioned grid counts
S9	`nlcsplit/step_i_gridrepr.py`, `evidence/recompute/20260929T_step_i_gridrepr.out`	the two discrepancies that were previously reported as one number
S10	`evidence/l2cp/20260928T213839Z_centos-step_h_l2cp.out`, `evidence/l2cp/20260929T051912Z_centos-l2_benzmeth_cp.out`, `nlcsplit/logs/step_h_l2cp.json`	level-2 with counterpoise, all six S22 subsystems complete, including the `p_over_ds` column behind the 0.92-1.08 ratio
S11	`RESEARCH-20260929-lit.md` §6.2-6.3	published comparison value for JCTC 3c00903 (full text via ChemRxiv green OA 10.26434/chemrxiv-2023-nc64t)
S12	`evidence/recompute/20260930T191014Z_stepg-official-basis` (official S22 geometry, the numbers quoted in sections 2 and 3.4), `evidence/recompute/20260929T003144Z_stepg-recompute.out` (constructed geometry, quoted as the cross-check), both produced by `nlcsplit/step_g_basis.py`, cross-checked against `evidence/centos/20260928T165857Z_centos-step2.out`	five-basis ladder of P_frz and of the four-scheme span at level 1, run once per geometry by the same script (`NCOFF` selects the official one). The two ladders agree to 5-8% on every span and to within 0.008 kcal/mol on the STO-3G-to-aug-cc-pVDZ drift, so the reading is geometry-insensitive; the script's own criterion (last-two-rung drift < 0.05) is satisfied on both geometries, which is what licenses calling the basis near-converged while the scheme span is not shrinking
S13	`evidence/centos/20260928T165857Z_centos-step2.out` lines 17-20, per-level values at lines 2-15	cross-grid spread per basis (sto-3g 0.0010, 6-31G* 0.0081, aug-cc-pVDZ 0.0067); the 0.0002 for 6-31G over level 1 to 2 is the difference of lines 8 and 9
S14	Lazić, Atodiresei, Alaei, Caciuc, Blügel and Brako, *Comput. Phys. Commun.* **181**, 371-379 (2010), DOI 10.1016/j.cpc.2009.09.016; open preprint arXiv:0810.2273v1; judgement in `RESEARCH-20260929-lit.md` §10	the closest published decomposition of a nonlocal term, read in full text
S15	`evidence/retrieval_calibration/oa_stab.js`, `evidence/retrieval_calibration/openalex_stability.json`, `evidence/retrieval_calibration/xref_calib2.js`, `evidence/retrieval_calibration/xref_r3.js`	measured recall of the open indexes: 4/4 known titles on rank 0, 0/4 and 1/3 for legitimate topical strings
S16	`RESEARCH-20260929-lit.md` §8.1	the institutional-access precondition of the search, and why the first round was counted from the UI
S17	`RESEARCH-20260929-lit.md` §13	second round: full enumeration through WoS's own record endpoint (RecordsSearched = 58,664,816, search date 2026-09-29), the three calibration strings reproduced record-for-record, the two zero-hit strings, complete readings of the two 4-record

freezing the monomers while switching only the dimer's convention printed -23.16 kcal/mol for the water dimer, which is impossible, and the impossibility was 17 kcal of oxygen 1s correlation counted on one side of the difference only. Two runs are kept because the second one exists to prove the first is still valid: after the frozen-core selection was refactored into a single call feeding both sides of the difference, the ladder was re-run with the script as it now ships, and stripped of wall-clock columns the two files differ in five lines, all of them one print statement whose wording changed - no energy moved. `nlcsplit/tests/test_cc_convention.py` pins that invariant, and one of its four tests rebuilds the broken pairing on purpose to show the guard can fail. `evidence/ccsdt/orca_crosscheck.py` is the third-program check on the same quantity: ORCA 6.0.1, same geometries, same basis, same counterpoise convention, on three systems and both conventions (eight rows, 24 single points), worst difference 0.0002 kcal/mol on the interaction energy and 1.2e-07 Ha between executions on the absolute energies. `evidence/ccsdt/20261001T052943Z_orca_crosscheck_three_systems.out` is the one to cite for it; it also carries the correction of an earlier claim in this directory, which is why the basis check is stated the way it is. The first version of that check read ORCA's `Number of basis functions` field against PySCF's orbital count and reported agreement at 18 for a water monomer in 6-31G - true, but not a criterion: for the same monomer in aug-cc-pVDZ the field prints 47 while ORCA's own SCF and coupled-cluster sections print `Basis Dimension Dim .... 41` and `Number of AO's ... 41`, which is PySCF's number. So the script now reads back the AO dimension per part and compares it with the PySCF nao for the same geometry and basis (24/24 equal), and its verdict refuses to print "agree" unless the energy test and the basis test hold at the same time. The 47 itself is recorded as unexplained rather than rationalised: 25 shells with maximum angular momentum 2 admits no s/p/d/sp bookkeeping that yields both 47 and 41. The frozen-core convention of each ORCA run is likewise read from its own output (`chemical core (4 el)` on a dimer, `(2 el)` on a monomer, `NO frozen core` on the all-electron rungs) rather than assumed from the input keyword |

Three absences are stated rather than hidden. `evidence/centos/...step_f_h2o2.out` predates the water-dimer geometry fix (its inter-fragment minimum is 2.0258 A against 1.95 A in `evidence/l2cp/`), so S8 cites it only for the component accounting, never as a current-geometry result. S15's files were sitting in `scratch/`, which is gitignored - they were promoted into `evidence/` on 2026-09-29 for exactly that reason. And one distinction has to be kept: S6 and S13 come from the same CentOS batch timestamped 2026-09-28T16:58, which is earlier than the geometry fix, while S12's primary artefact is a recompute run afterwards. Within that earlier batch the initial S22 structure and the relaxed one are distinguishable (the level-0 6-31G* value -0.5316 appears in S13's ladder, while the -0.5975 of the relaxed structure does not), so S13's cross-grid spreads are read here as measured on the *initial* S22 geometry, not on the relaxed one that carries the headline numbers. The level-1 column of that same ladder agrees with the later recompute to four decimals for all three bases the two runs share, so the grid-convergence conclusion does not turn on which geometry was used - but that was checked after the fact. The full ladder has since been re-run on the geometry that carries the headline numbers, and section 3.4 quotes that run (`evidence/recompute/20260930T191014Z_stepg-official.out`); what remains true is the weaker thing: the cross-grid spreads of S13 are still measured on the initial structure, because that is the batch which contains all four grid levels.

References

[1] Oleg A. Vydrov and Troy Van Voorhis. Nonlocal van der waals density functional made simple. *Physical Review Letters*, 103:063004, 2009. doi: 10.1103/PhysRevLett.103.063004.

[2] Narbe Mardirossian and Martin Head-Gordon. ωb97x-v: A 10-parameter, range-separated hybrid, generalized gradient approximation density functional with nonlocal correlation, designed by a survival-of-the-fittest strategy. *Physical Chemistry Chemical Physics*, 16:9904, 2014. doi: 10.1039/c3cp54374a.

[3] Narbe Mardirossian and Martin Head-Gordon. ω b97m-v: A combinatorially optimized, range-separated hybrid, meta-gga density functional with vv10 nonlocal correlation. *The Journal of Chemical Physics*, 144:214110, 2016. doi: 10.1063/1.4952647.

[4] Joaquín Calbo, Enrique Ortí, Juan C. Sancho-García, and Juan Aragó. The nonlocal correlation density functional vv10. *Annual Reports in Computational Chemistry*, 11:37–102, 2015. doi: 10.1016/bs.arcc.2015.09.002.

[5] Susi Lehtola, Conrad Steigemann, Micael J. T. Oliveira, and Miguel A. L. Marques. Recent developments in libxc — A comprehensive library of functionals for density functional theory. *SoftwareX*, 7:1–5, 2018. doi: 10.1016/j.softx.2017.11.002.

[6] Saswata Dasgupta, Etienne Palos, Yuanhui Pan, and Francesco Paesani. Balance between physical interpretability and energetic predictability in widely used dispersion-corrected density functionals. *Journal of Chemical Theory and Computation*, 20:49–67, 2023. doi: 10.1021/acs.jctc.3c00903.

[7] Andreas Heßelmann. Assessment of a nonlocal correction scheme to semilocal density functional theory methods. *Journal of Chemical Theory and Computation*, 9:273–283, 2013. doi: 10.1021/ct300735g.

[8] Predrag Lazić, Nicolae Atodiresei, Mojtaba Alaei, Vasile Caciuc, Stefan Blügel, and Radovan Brako. Junolo – jülich nonlocal code for parallel post-processing evaluation of vdw-df correlation energy. *Computer Physics Communications*, 181:371–379, 2010. doi: 10.1016/j.cpc.2009.09.016.

[9] Per Hyldgaard, Kristian Berland, and Elsebeth Schröder. Interpretation of van der waals density functionals. *Physical Review B*, 90:075148, 2014. doi: 10.1103/PhysRevB.90.075148.

[10] Holger Kruse, Pavel Banáš, and Jiří Šponer. Investigations of stacked dna base-pair steps: Highly accurate stacking interaction energies, energy decomposition, and many-body stacking effects. *Journal of Chemical Theory and Computation*, 15:95–115, 2018. doi: 10.1021/acs.jctc.8b00643.

[11] Waldemar Hujo and Stefan Grimme. Comparison of the performance of dispersion-corrected density functional theory for weak hydrogen bonds. *Physical Chemistry Chemical Physics*, 13:13942, 2011. doi: 10.1039/c1cp20591a.

[12] Debalina Sinha and Michele Pavanello. Exact kinetic energy enables accurate evaluation of weak interactions by the fde-vdw method. *The Journal of Chemical Physics*, 143:084120, 2015. doi: 10.1063/1.4928531.

[13] Ikutaro Hamada and Masaru Tsukada. Adsorption of C_{60} on Au(111) revisited: A van der waals density functional study. *Physical Review B*, 83:245437, 2011. doi: 10.1103/physrevb.83.245437.

[14] Oleg A. Vydrov and Troy Van Voorhis. Nonlocal van der waals density functional: The simpler the better. *The Journal of Chemical Physics*, 133:244103, 2010. doi: 10.1063/1.3521275.

[15] M. Dion, H. Rydberg, E. Schröder, D. C. Langreth, and B. I. Lundqvist. Van der waals density functional for general geometries. *Physical Review Letters*, 92:246401, 2004. doi: 10.1103/PhysRevLett.92.246401.

[16] A. D. Becke. A multicenter numerical integration scheme for polyatomic molecules. *The Journal of Chemical Physics*, 88:2547–2553, 1988. doi: 10.1063/1.454033.

[17] R.Eric Stratmann, Gustavo E. Scuseria, and Michael J. Frisch. Achieving linear scaling in exchange-correlation density functional quadratures. *Chemical Physics Letters*, 257:213–223, 1996. doi: 10.1016/0009-2614(96)00600-8.

[18] Lars Goerigk, Andreas Hansen, Christoph Bauer, Stephan Ehrlich, Asim Najibi, and Stefan Grimme. A look at the density functional theory zoo with the advanced gmtkn55 database for general main group thermochemistry, kinetics and noncovalent interactions. *Physical Chemistry Chemical Physics*, 19:32184–32215, 2017. doi: 10.1039/c7cp04913g.

[19] Petr Jurečka, Jiří Šponer, Jiří Černý, and Pavel Hobza. Benchmark database of accurate (MP2 and CCSD(T) complete basis set limit) interaction energies of small model complexes, DNA base pairs, and amino acid pairs. *Physical Chemistry Chemical Physics*, 8:1985–1993, 2006. doi: 10.1039/b600027d.

[20] Zekeriya Altun, Frank Neese, and Agnieszka Bistoni. Local energy decomposition analysis of hydrogen-bonded dimers within a domain-based pair natural orbital coupled cluster study. *Beilstein Journal of Organic Chemistry*, 14:919–929, 2018. doi: 10.3762/bjoc.14.79.

[21] Taku Takatani, Edward G. Hohenstein, Krzysztof Malagoli, Michael W. Marshall, and Charles D. Sherrill. Basis set consistent revision of the S22 test set of noncovalent interaction energies. *The Journal of Chemical Physics*, 132:144104, 2010. doi: 10.1063/1.3378024.

[22] Qiming Sun, Timothy C. Berkelbach, Nick S. Blunt, George H. Booth, Sheng Guo, Zhendong Li, Junzi Liu, James D. McClain, Elvira R. Sayfutyarova, Sandeep Sharma, Sebastian Wouters, and Garnet Kin-Lic Chan. PySCF: the Python-based simulations of chemistry framework. *WIREs Computational Molecular Science*, 8:e1340, 2018. doi: 10.1002/wcms.1340.

[23] Qiming Sun, Xing Zhang, Samragni Banerjee, Peng Bao, Marc Barbry, Nick S. Blunt, Nikolay A. Bogdanov, George H. Booth, Jia Chen, Zhi-Hao Cui, Janus J. Eriksen, Yang Gao, Sheng Guo, Jan Hermann, Matthew R. Hermes, Kevin Koh, Peter Koval, Susi Lehtola, Zhendong Li, Junzi Liu, Narbe Mardirossian, James D. McClain, Mario Motta, Bastien Mussard, Hung Q. Pham, Artem Pulkin, Wirawan Purwanto, Paul J. Robinson, Enrico Ronca, Elvira R. Sayfutyarova, Maximilian Scheurer, Henry F. Schurkus, James E. T. Smith, Chong Sun, Shi-Ning Sun, Shiv Upadhyay, Lucas K. Wagner, Xiao Wang, Alec White, James Daniel Whitfield, Mark J. Williamson, Sebastian Wouters, Jun Yang, Jason M. Yu, Tianyu Zhu, Timothy C. Berkelbach, Sandeep Sharma, Alexander Yu. Sokolov, and Garnet Kin-Lic Chan. Recent developments in the PySCF program package. *The Journal of Chemical Physics*, 153: 024109, 2020. doi: 10.1063/5.0006074.

[24] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi:

10.1038/s41586-020-2649-2.